Optimizing Inventory Management and Strategizing Expansion of Corindo Spices

Final Report for the BDM Capstone Project

KHUSHI PANDEY

23f1002421

IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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1. Executive Summary

Corindo Spices is a regional wholesale spice business situated in Rajbandigarh, Gouripur, West Bengal, with aspirations to expand its market presence across the state. The organization faces two primary challenges: inefficient inventory management during the launch of new products and a lack of structured strategies for regional market expansion amid established competition.

To address these concerns, a comprehensive data-driven analysis was conducted using 12 months of order and sales data, collected from April 2024 to March 2025. The dataset encompassed information from ten districts, eight spice products, and three distinct pack sizes. Statistical techniques and quantitative methods such as ABC classification, fulfilment rate analysis, district-wise sales analysis, moving average forecasting, and analogous product analysis were employed using Python and Microsoft Excel.

The analysis revealed that Poppy Seeds, Panch Phoron, and Turmeric serve as the primary revenue drivers, particularly during festive seasons. Quality control issues were discovered in Turmeric and Chilli products that led to a temporary decline in sales in late 2024, which recovered following improved packaging. District-level analysis indicated that North 24 Parganas and Hooghly are high-potential regions for expansion due to consistent demand and strong fulfilment rates.

The report concludes that improved marketing, ABC-driven inventory control, and targeted regional strategies can enhance Corindo Spices' operational efficiency, optimize stock levels, and support sustainable growth across new markets.

2. Explanation of Analysis Methods:

2.1. Data Collection

Data was gathered through primary collaboration with Corindo Spices, capturing monthly order and sales data for the period April 2024–March 2025. It included 8 spice products across 10 districts and 3 pack sizes, providing a view of product performance across geographies and packaging segments.

This dataset served as the foundation for the analyses addressing the two main problem statements:

1. Inventory Management: determining optimal stock levels and identifying key revenue drivers.

2. Regional Expansion: pinpointing strong-performing districts for market penetration.

2.2 Data Cleaning and Pre-processing

Data cleaning was a crucial initial step to ensure analytical reliability. Several inconsistencies were identified and corrected:

- *Pack size mislabeling*: Instances of "5mrp" incorrectly entered as "10mrp" were standardized after owner verification.

- *Sales greater than orders*: Some rows contained unrealistic entries (sales > orders). These were corrected through owner consultation.

- *Data restructuring*: Each month's data was transformed from a wide format to a long format using Python's Pandas library to facilitate easier aggregation and visualization.

Data cleaning ensures high-quality analysis by:

- Eliminating inconsistencies that could skew summary statistics.

- Maintaining uniform units, scales, and labels across datasets.
- Enhancing the reliability of statistical and forecasting results.

2.3 Quantitative Analysis Methods

The study utilized five core analytical methods to interpret and address the company's operational challenges.

2.3.1 ABC Classification

Tool Used: Microsoft Excel

Explanation:

ABC analysis classifies inventory into three categories based on their cumulative contribution to total sales.

$$\text{Cumulative Percentage} = \frac{\text{Cumulative Sales}}{\text{Total Sales}} \times 100$$

- Category A: Top 70% of sales value (critical items).
- Category B: Next 20% (moderately significant).
- Category C: Remaining 10% (low-value, high-volume items).

This classification enables Corindo Spices to prioritize inventory decisions—allocating more resources toward Category A items that significantly influence revenue and customer satisfaction.

2.3.2 Fulfilment Rate Analysis

Tool Used: Microsoft Excel

Explanation:

The fulfilment rate quantifies the efficiency of the supply chain, showing how effectively customer demand is met.

$$\text{Fulfilment Rate} = \frac{\text{Sales}}{\text{Orders}} \times 100$$

A high fulfilment rate signifies operational efficiency and reliable distribution. Conversely, lower rates point to bottlenecks in logistics or insufficient inventory.

For Corindo Spices, identifying districts with higher fulfilment rates helps guide strategic expansion into areas already exhibiting logistical robustness.

2.3.3 District-wise Sales Analysis

Tool Used: Microsoft Excel

Explanation:

District-wise aggregation of sales data helps identify regions contributing most significantly to overall revenue.

Regional sales distribution supports targeted marketing and expansion efforts by identifying strong-performing districts that can serve as anchors for further market penetration.

2.3.4 Moving Average Analysis

Tool Used: Pandas, Python

Explanation:

The moving average technique was applied to smooth short-term fluctuations in order and sales data. A 3-month moving average was chosen as it provides a balanced reflection of seasonal variations without overreacting to short-term noise.

Mathematically,

$$\mathrm{MA}_t = \frac{X_t + X_{t-1} + X_{t-2}}{3}$$

where $X_t$denotes sales in month t.

The moving average, for Corindo Spices, helps determine appropriate stock levels, minimize holding costs, and anticipate seasonal surges. The 3-month window aligns with typical quarterly business cycles and production lead times.

2.3.5 Analogous Product Analysis

Tool Used: Python and Microsoft Excel

Explanation:
This method estimates potential demand for upcoming products—Shahi Garam Masala, Kashmiri Lal Mirch, and Coriander Powder—by identifying similar (analogous) products with comparable characteristics such as pricing, customer profile, and sales trends.

Assuming Corindo initially captures 20% of the existing market share for analogous spices:

$$\text{Expected Orders} = (\text{Average Monthly Sales of Analogous Product}) \times 0.2$$

To ensure availability, a safety stock margin is added:

$$\text{Initial Stock} = \text{Expected Orders} \times (1 + \text{Safety Stock }\%)$$

Analogous product analysis combines empirical data and business reasoning to estimate feasible launch volumes and pricing. It provides realistic forecasts based on market benchmarks and minimizes risk during new product introduction.

3. Results and Findings:

ABC Analysis:

Cumulative % <70%=A, 70%-90%=B, 90%-100%=C

Product	Order Quantity	% of Total	Cumulative %	Category
Poppy	106034	22.0905	22.0905087	A
Panch Phoran	97461	20.3045	42.3949683	A
Turmeric	85856	17.8867	60.2817095	A
Cumin	58222	12.1296	72.4113434	B
Chilli	39055	8.13649	80.5478356	B
Yellow Mustarc	37350	7.78128	88.329118	B
Brown Mustarc	31318	6.52461	94.8537286	C
Fennel	24702	5.14627	100	C
Total	479998			

The ABC classification revealed the following categories:

- **Category A (High Priority):** Poppy Seeds, Panch Phoron, Turmeric
- **Category B (Moderate Priority):** Cumin, Chilli, Yellow Mustard
- **Category C (Low Priority):** Brown Mustard, Fennel Seeds

Category A products deserve maximum attention for procurement, marketing, and quality control. These are the company's "bread and butter" spices, and stockouts could directly impact profitability.

Category B products, while moderately important, should maintain balanced stock levels to ensure availability without excessive investment.

Category C products, with minimal contribution, may be produced in smaller quantities to free up storage and capital resources.

Average Fulfilment Rate of all Spices across 10 Districts:

	Average Fulfilment Rate							
Row Labels	**Poppy**	**Cumin**	**Brown**	**Yellow**	**Fennel**	**Panch**	**Turme**	**Chilli**
BANKURA	99.4	97.74	93.26	99.58	98.7	98.7	98.4	100
BIRBHUM	98.6	98.41	96.47	98.61	97.5	99.1	98.7	99
HOOGHLY	97.7	98.91	97.2	99.71	97.8	99.1	99.7	100
KOLKATA	99.9	98.55	92.1	98.43	98.1	98.6	99.6	99
MURSHIDABAD	98.8	97.78	95.04	98.29	98.8	98.1	99.5	99
NADIA	99.4	98.37	95.71	98.48	98.5	98.2	99.3	98
NORTH 24 PGS.	99.1	98.78	97.39	99.04	98.2	99.5	99.4	100
PASCHIM BURDWAN	98.3	98.52	96.05	98.77	97.5	98.5	99.5	100
PASCHIM MIDNAPORE	98	97.87	95.89	97.93	98.4	99.3	82.9	83
PURBA BURDWAN	99.1	98.56	95.96	98.63	97.8	99	99.5	100

The average fulfilment rate across all districts exceeded 90%, signifying a strong order execution system. Minor deviations were observed in:

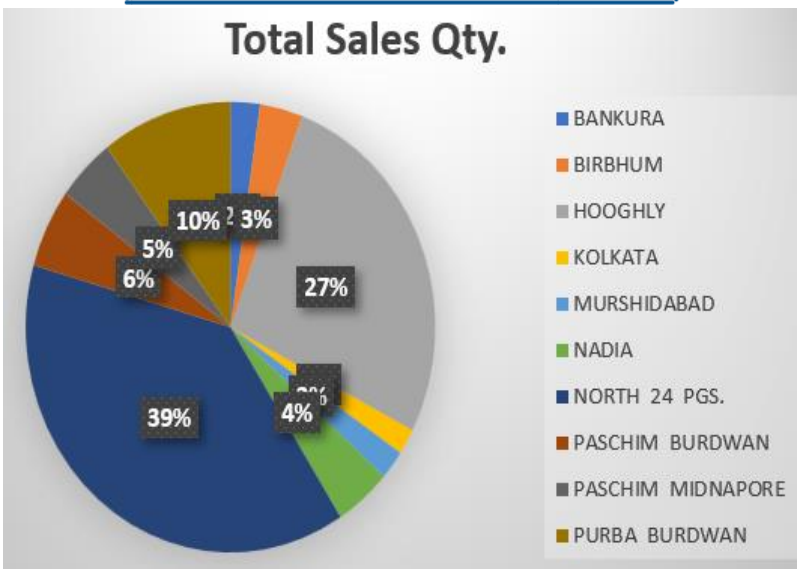
- **Turmeric:** 82.9% (Paschim Midnapore)
- **Chilli:** 82.6% (Paschim Midnapore)

The consistently high fulfilment rate demonstrates that Corindo Spices has a dependable supply chain with adequate distribution coverage. However, the lower rates observed for Turmeric and Chilli indicate product-specific weaknesses, mainly packaging and storage limitations, that can disrupt service quality. Discussions with the business owner revealed temporary distribution gaps caused by delayed shipments.

Overall, the fulfilment rate percentage remained above 90%, satisfying the business's service-level targets and implying readiness for expansion without extensive infrastructural overhaul.

Number of Products (inclusive of all pack sizes) sold across 10 Districts:

Total Sales Qty(from April'24-March'25)	
NORTH 24 PGS.	182456
HOOGHLY	127422
PURBA BURDWAN	49167
PASCHIM BURDWAN	26814
PASCHIM MIDNAPORE	21851
NADIA	20935
BIRBHUM	16084
BANKURA	11004
MURSHIDABAD	10088
KOLKATA	8729



- **North 24 Parganas:** ≈ 39 % of total sales - serves as the primary distribution hub, supported by dense retail networks and proximity to Kolkata.

- **Hooghly:** ≈ 27 % of total sales - second-largest contributor with steady growth in bulk (500 g) orders.

- **Paschim Midnapore & Purba Burdwan:** ≈ 16 % combined - emerging growth zones for household packs.

- **Remaining districts:** low penetration but potential for gradual expansion through wholesalers.

The analysis thus defines a *two-tier market structure*: a mature primary market (North 24 Parganas and Hooghly) and secondary markets that require targeted marketing and dealer support.

Analogous Product Analysis

Market prices of comparable brands (JK, Sunrise, Ganesh) were studied for 50 g packs

Corindo Spices Ideal Price Suggestion (50g pack)

Product	Min Price	Max Price	Average Price	Ideal_Competitive_Price
Shahi Garam Masala	₹ 90.00	₹ 130.00	₹ 116.67	₹ 105.00
Kashmiri Lal Mirch	₹ 60.00	₹ 70.00	₹ 66.67	₹ 60.00
Coriander Powder	₹ 20.00	₹ 20.00	₹ 20.00	₹ 18.00

Findings include:

- **Shahi Garam Masala:** Exhibited the highest price variation, suggesting a differentiated product positioning opportunity.
- **Coriander Powder:** Highly commoditized—uniform pricing across brands, requiring competitive differentiation through quality and packaging.
- **Kashmiri Lal Mirch:** Positioned in the premium segment with relatively stable pricing.

Estimated Demand:
Based on analogous products (Chilli and Cumin powders), the expected monthly order volumes were estimated at 404 and 520 units, respectively.

Assuming that ABC Spices will initially capture around 20% of the existing market for similar spices:

$$\textit{Expected Orders} = \textit{Average Monthly Sales of Analogous Product} \times 0.2$$

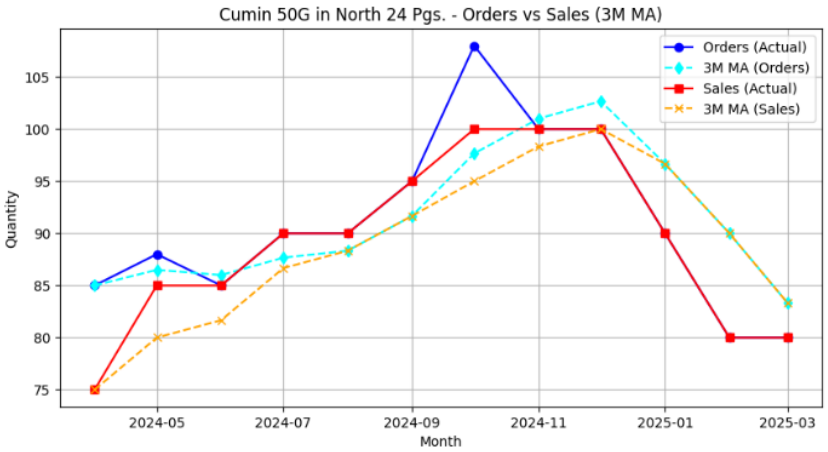
$$\textit{Initial Stock} = \textit{Expected Orders} \times (1 + \textit{Safety Stock \%})$$

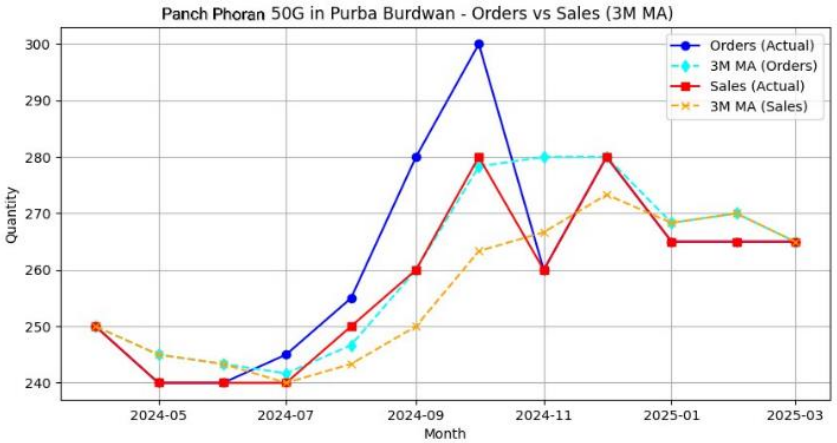
Product	Expected Orders	Safety Stock	Initial Stock
Shahi Garam Masala	81	20%	97
Kashmiri Lal Mirch	81	20%	97
Coriander Powder	104	20%	125

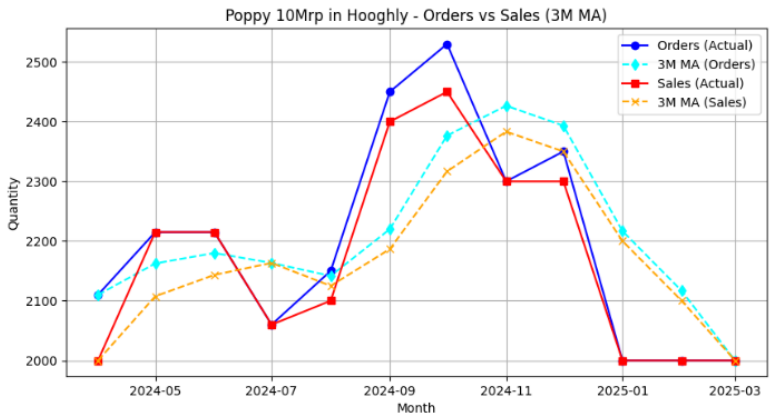
Applying the 20% market capture rule, Corindo's initial monthly expected orders are forecasted at approximately 80–100 units per new product, with an additional 20% safety stock margin.

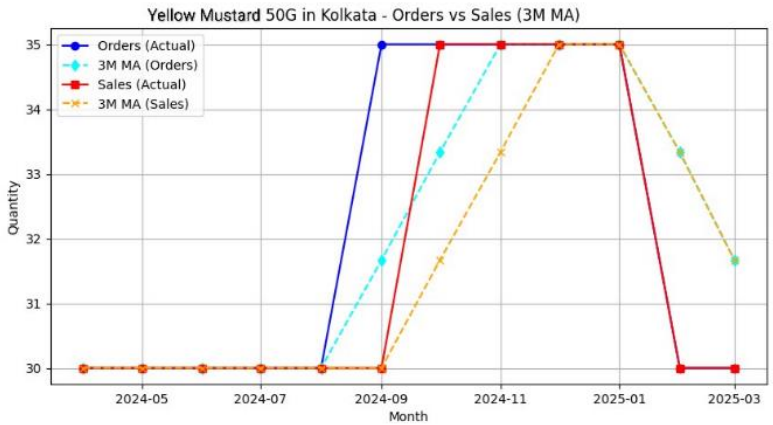
The analogous analysis offered quantitative insight into market entry strategy, assisting Corindo Spices in realistic production planning and competitive pricing, reducing the risk of overproduction or understocking during product introduction.

Moving Average Analysis



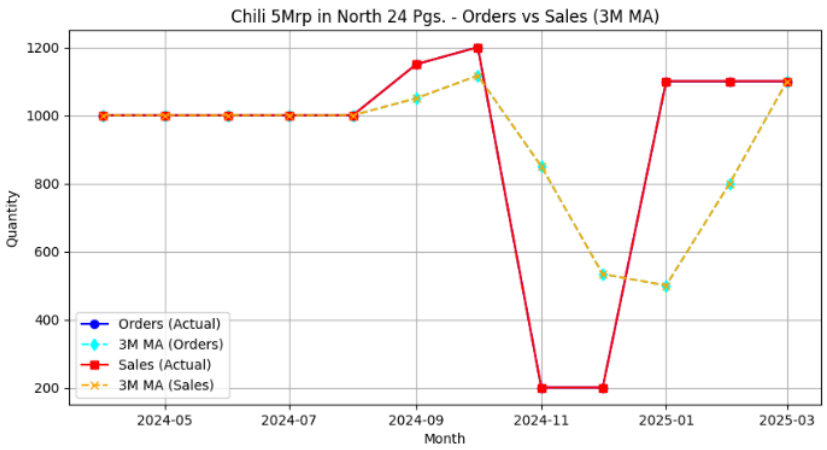


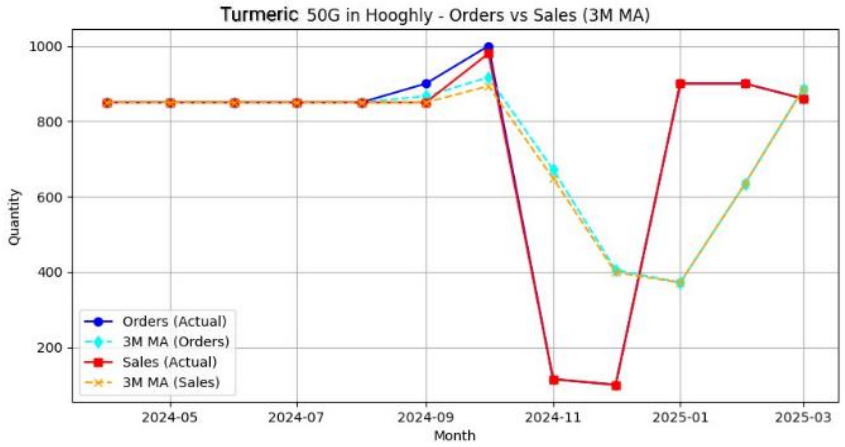




The 3-month moving averages of sales for all spices revealed consistent seasonal patterns:

- Poppy and Fennel**:** Show peak sales during summer months (April–June).
  Poppy : 7950 [May'24] vs 7355 [Jan'25]
- Turmeric and Chilli: Experience demand surges during festive seasons (September–November).
- Brown Mustard and Yellow Mustard: Exhibit steady demand year-round with minimal fluctuations.





An anomaly was detected for **Chilli** and **Turmeric** during **November–December 2024**, where both experienced sharp declines in sales.

- *Meeting with the owner*- Upon discussion with the business owner, it was revealed that in October, quality issues were reported for both Turmeric and Chilli Powder, which subsequently led to decreased orders across several districts. The products were noted to become moist and lose their freshness (likely due to poor packaging).
  The impact was particularly sharp in Hooghly, North 24 Parganas, Paschim Midnapore, and Purba Burdwan for Turmeric, and in Hooghly, North 24 Parganas, and Purba Burdwan for Chilli.
  The company then approached a more secured and suitable packing for these two spices and reassured the wholesalers and retailers of the standard quality of spices.

	Turmeric		Chilli	
	Orders	Sales	Orders	Sales
Oct'24	8946	8521	3980	3900
Nov'24	3990	3990	2380	2380
Dec'24	3655	3655	2380	2380
Jan'24	7945	7945	3500	3500

Once packaging materials were replaced and supplier communication improved, sales rebounded by January 2025.

Demand for Turmeric and Chilli rebounded by ≈ 217 % and 147 % respectively, post intervention, demonstrating the impact of operational corrective actions.

4. Interpretation of Results and Recommendations

4.1 Interpretation

The analyses jointly demonstrate the following:

- **Inventory Optimization:**
  The ABC analysis highlights an unbalanced focus on lower-performing products. Products in Category A (Poppy Seeds, Panch Phoron, Turmeric) should receive high attention in stocking, supply chain management due to their significant impact on sales. Category B products (Cumin, Chilli, Yellow Mustard) do not require as much attention as A but demand monitoring can help avoid overstocking of these products.

- **Operational Efficiency:**
  High fulfilment rates validate the existing logistics model. However, small deviations in select districts indicate the need for periodic monitoring to ensure consistent delivery performance.

- **Seasonality and Forecasting:**
  Moving average trends confirm that demand is season-sensitive. Seasonal forecasting models should be institutionalized to predict demand three months in advance.

- **Quality Assurance:**
  The dip in Turmeric and Chilli sales underscores the importance of quality control and packaging integrity. Product consistency directly correlates with sales stability and customer trust.
- **Market Readiness:**
  District-level analysis underscores North 24 Parganas and Hooghly as core regions to pilot new launches. Their established wholesale networks can serve as expansion anchors. High-performing districts can act as strategic hubs for expansion. Targeting these regions first ensures better logistics, cost optimization, and customer awareness. Expansion into adjacent districts can be achieved later through a hub-and-spoke model of distribution.

4.2 Recommendations

Based on these interpretations, the following actionable recommendations are proposed:

A. Seasonal Demand Planning

Action Plan:

- Scale up Poppy & Fennel inventory in March-July; increase others in September-December.
- Run festive discounts or combo offers during Durga Puja-Diwali season.
- Maintain a minimum safety stock of 10–15% for Category A products.

B. Market Expansion Beyond Core Districts

Action Plan:

- Prioritize Paschim Burdwan, Nadia, and Paschim Midnapore for expansion (higher literacy & moderate income base).
- Introduce micro-distributors or rural stock points for Bankura and Birbhum where access is low.
- Offer introductory combo packs (e.g., "Festive Spice Kit") to boost household trials.

- Utilize existing wholesale channels to introduce bulk packs in emerging districts.

C. Pricing & Product Strategy

Action Plan:

- Maintain competitive pricing slightly below market average as per analysis
  - *Shahi garam Masala* *Rs. 105*
  - *Kashmiri Lal Mirch* *Rs. 60*
  - *Coriander Powder* *Rs. 18*
- Leverage successful districts as pilot markets for new product introductions.

D. Product Quality and Packaging

Action Plan:

- Continue using moisture-resistant packaging for all spices, especially Chilli and Turmeric.
- Introduce periodic product audits to ensure freshness and consistent quality.
- Display QR-coded quality assurance labels on packaging to build brand credibility.

E. Sales and Marketing

Action Plan:

- Digital outreach: Promote through regional social media channels emphasizing authentic Bengali cooking.
- Sampling campaigns: Distribute sample sachets of premium blends in local fairs, and retail stores.
- Tie-ups with local eateries for "powered by Corindo Spices" branding to build trust in smaller districts.

F. Data and Performance Monitoring

Action Plan:

- Implement monthly district-wise dashboards tracking:
  - ✓ Sales growth rate
  - ✓ Fulfillment rate
  - ✓ Stock-out frequency
  - ✓ Consumer returns or complaints

  Use this to iteratively tweak pricing and marketing per region.

5. Conclusion

The comprehensive data-driven evaluation of Corindo Spices reveals that strategic inventory management and targeted regional expansion can significantly enhance the company's market performance. Through ABC analysis, fulfilment rate assessments, and moving average analysis, the study identifies core strengths, high operational efficiency and stable demand in specific districts, as well as improvement areas, such as packaging consistency and product prioritization.

By adopting the recommendations presented, Corindo Spices can transform from a regionally concentrated business into a scalable enterprise with structured inventory control, predictive demand planning, and high-quality product standards.

The success of this project demonstrates the importance of combining data analytics with business insight.

Overall, this analytical framework provides a strong foundation for Corindo Spices to achieve sustained profitability and successful state-wide expansion.